

Communication blackouts in power outages: Findings from scenario exercises in Germany and France

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ABSTRACT

Although infrequent, power outages can cause major disruption and incur huge costs. The increasing demand for electricity, more extreme weather events and the possibility of cyber-attack are increasing the risk of blackouts. This paper reports using scenario planning and role-playing exercises with German and French university students to investigate people's possible reactions and their expectations of government during a hypothetical 3-day blackout. The scenario is designed to explore how society would cope with a power failure over successive time periods. The aim is provide information to emergency managers and policy makers about community needs and people's likely behavior in future blackouts.

1. Introduction

The chance of a long lasting power blackout is low especially in the highly industrialized societies like Europe, but in today's strongly interconnected world even a small disruption in a power network can escalate to a large disruption and systemic failure of critical infrastructure [1]. In the US and Europe the likelihood of large-area failures seems to be greater than might be expected on the basis of extrapolation from small failures [2]. The risk of power blackout is increasing due to technical failure; natural events, cyber-attacks, human failure during maintenance work or a combination of factors [3,4]. Power outages in Italy in 2003, southwest Europe in 2006 and Turkey in 2015 are the examples of blackouts in Europe that caused serious disruptions, despite lasting less than 24 h [5]. Operational reliability of the power system is closely linked to the security of telecommunication networks, including its ability to withstand cyber-attacks [6]. Improving operational reliability is currently a major concern of governments of many countries, including Germany and France. Although infrequent, extensive power outages can cause major disruption and incur huge costs. European power networks extend across country borders and cross border blackouts pose particular problems for inter-agency communication and cooperation. The aim of this research is to explore the potential risks and impacts of a major blackout and to make recommendations for any

changes to current policies and procedures.

A role-play scenario planning with university students is used to explore societal resilience to a hypothetical 72 h power failure in Germany and France. The objective was to examine how people's behavior, emotions and reactions change over the 3 days and to see to what extent they are likely to cooperate with family, friends and neighbors and fend for themselves and how their faith and trust in the authorities and agencies fluctuates.

1.1. Human response to emergencies

There has been extensive research into people's reactions in crisis situations going back to Prince's [7] study of a ship collision in 1917. Drabek and Evans [8] summarised the major contributions of American sociologists to an understanding of human responses to disaster. The first issue relates to labelling panic, looting and other anti-social behavior as myths [9–15]. Quarantelli [16] suggested that people as a whole do not panic and rather than reacting in an anti-social manner, people typically become more cohesive and unified during situations of collective stress. Tierney [17] writes that the incidence of panic behavior is vanishingly rare in crisis situations and despite fear and the entirely reasonable desire to flee, people in severe danger try to make rational decisions and work collectively. In an analysis of the World

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Trade Centre disaster 2001 extreme behavioral action, including panic, was noted in less than 1% of cases [18]. [19] suggest that anti-social behaviors during disaster scenarios, is, at best, over-exaggerated and that human behavior in disaster scenarios can only frequently be described as “pro-social”. In contrast to panic behavior, the typical human response to disaster warnings is disbelief, denial, and reinterpretation to reduce or eliminate the threat potential [20].

On the other hand, there are examples of looting and crime in a small minority of disasters such as New York blackout 1977 [21,22], Hurricane Katrina [23]. Despite widespread fears of looting during the Kista blackout in a suburb of Stockholm, Sweden there was slightly more criminal activity than usual [24].

The second issue of relevance is that of emergent behavior – emergent groups of people working together in pursuit of collective goals relevant to the disaster but whose organization has yet to become institutionalized. Emergent phenomena have been a prevalent feature in disasters [25] and may on occasion lead to conflict with formal emergency management [26]. This is one aspect of the extensive literature on social resilience and adaptive behavior. Hutter et al. [27]; introducing a special edition on resilience, conclude that there are a multiplicity of perspectives on resilience and the lack of a clear definition poses a challenge to operationalising its use in disaster management. This is perhaps not surprising since resilience is a multi-dimensional concept encompassing social, economic, physical, technological and natural dimensions [28]. In social science, resilience is considered as a complex web of social connections and capacities that enable a community to cope with hazards [29,30]. Obrist [31] says resilience is a dynamic process, not as a static state, and social resilience enables people not only to cope with adverse conditions but also to create options and responds proactively to adversity. Keck and Sakdapolrak [32] also suggest that resilience goes beyond coping and encompasses adaptive and transformation processes.

Rogers and Nehnevajsa [33] report on behavior and attitudes under crisis conditions and suggest that people respond to crises in remarkably similar ways. The disruption created by disasters creates stress, yet individuals and organizations seek to maintain the continuity of behavior. The first response to impending danger reflects a strong commitment to protecting themselves, their loved ones and even others with whom they may become associated. The second type of response is information seeking. As people become alerted to the potential for danger they tend to seek information about the hazard, from their own senses and to get confirmation from others, particularly family and friends. Finally, there is a propensity to respond in groups, particularly families, by consolidating resources. These three basic behavior patterns were observed in the scenario workshops reported in this paper.

1.2. Power outages

Electricity is an enabling technology for a host of quotidian activities [34], so its disruption affects normal daily life and has a cascading effect on other infrastructure, for example water supply [35]. Electric power systems are part of the critical infrastructure of modern societies,

therefore is essential to boost their resilience [36]. Electricity infrastructure is necessary to sustain human and economic wellbeing [37]. Many countries have, therefore, introduced mandatory requirements for operators to maintain high reliability and stability of supply, for example, The U.S. Department of Energy, Federal Energy Regulatory Commission (FERC) [38] is working with power system operators to ensure reliability of the U.S. power system. In Germany, the transition from fossil fuels to renewables is pushing the industry towards greater flexibility and resilience through various innovations in market design and pricing mechanisms [39]. In France, RTE, the electricity transmission operator, introduced emergency response arrangements in the wake of the storms in December 1999. It is tackling current reliability challenges by leveraging power generation flexibility, stronger cooperation between transmission system operators and coordination centers and adherence to European grid codes [6]. However, power systems are highly nonlinear and it is challenging and uneconomical to make power systems totally resilient [40].

Drawing on examples from around the world, Wood et al. [41] argue that a crucial factor in people ability to cope with adversity is social capital: the extent to which social networks are able to support collective action, and promote psychological wellbeing. One of the earliest detailed accounts of people's reactions comes from the blackout across the eastern seaboard of the United States on 9 November 1965, affecting 25 million people. Six days after the breakdown, the National Opinion Research Centre (NORC) interviewed over a thousand people about how their reactions to the blackout. They found that most people were not in the least alarmed, there was no panic and the majority of people reported that strangers were unusually helpful and friendly to each other. Above all, they found that the blackout affected rhythms of communication and social interaction. For example, the New York Telephone Company reported most calls were to relatives and instead of turning for information to City Hall or formal organizations, people turned to neighbors and to local leaders in the neighborhood [42].

On August 14, 2003 a massive power outage caused blackouts throughout most of Ontario, along with parts of the US. This power failure was the largest in North American history, spanning 24,086 square kilometres and costing the Canadian economy \$1–2 billion [43]. Brenda Murphy [44] conducted a survey of 1200 Ontario residents six months after the outage to discover how they had managed to deal with the emergency. She found that families relied on networks of family, friends, neighbors and community associations to help them get through a crisis. 89% of respondents stated that their neighbors would provide assistance and 37% said they provided assistance during the blackout.

Although rare, Europe has experienced major power failures and maintaining stability of power supply is especially challenging during natural catastrophes [45]. In France and Germany storms and floods are the natural hazards cause the biggest problems. On September 23, 2003, nearly four million customers lost power in eastern Denmark and southern Sweden following a cascading outage that struck Scandinavia [46]. Days later, a cascading outage between Italy and the rest of central Europe left most of Italy in darkness on September 28 [47]. In France and Germany, there have been 10 serious examples of storms causing

Table 1

Power outages in France and Germany following storm events.

Storm	Date	Impacts of power outage	Reference
Lothar and Martin	Dec 1999	Extensive damage to the French and German national power grids	[48]
Dagmar	Dec 2004	Northern France, including Paris, affected leaving thousands of homes without power.	[49]
Renate	Oct 2006	SW coast of France battered leaving many homes without power for many hours	[50]
Klaus	Jan 2009	Over a million homes in southwest France lost power.	[51]
Katia	Sep 2011	Hurricane caused widespread power outages throughout Europe, including in France and Germany	[52]
Joachin	Dec 2011	Power outages in France and Germany	[53]
Niklas	Mar 2015	Power outages in Germany	[54]
Fabien	Dec 2019	Wind and storms cut power supply to 220,000 homes in southwestern France	[55]
Ciara	Feb 2020	Power cut to 130,000 homes in northern France	[56]
Sabine	Feb 2020	60,000 homes were left without electricity in Bavaria	[57]

temporary power outages in the past 10 years (Table 1).

Electricity outages can be both an effect and a cause of cascading effects [58]. For example, the 2013 Germany floods caused widespread disruptions in power supply, which in turn affected water treatment plants and public transport amongst other services and activities [59]. In February 2014 several storms again affected UK and 80,000 homes were left without power for several days. The impacts of power outages on households, the description of the challenges faced and the strategies adopted by them are described by Ghanem et al. [60]. The capacity of households to deal with power outages depends not only on the household itself, but also on the social resilience of the whole community. Cascading effects are common in disaster events and a chain of interaction can amplify the impact on society where the initial power failure can lead to fires, an increase in accidents and fatalities, stresses to the health sector, water shortages and hygiene issues, disruption of refrigeration and food shortages, environmental contamination, carbon monoxide poisoning from disruption of underground ventilation, traffic disruption, disruption to temperature control, people becoming trapped, increased crime, stress on police and emergency services, business continuity and logistics failure and last, but not least communications system and telecoms failure [61].

The 2011 Tohoku earthquake, in particular to the Fukushima Daiichi nuclear disaster, was a wake-up call to the risk of “cascading disasters” [62]. A paper by Pescaroli and Alexander [63] addresses the vulnerability of critical infrastructure to cascading effects of disasters. They suggest a shift attention from risk scenarios based on hazard to vulnerability scenarios based on potential cascading effects in which they identify nodes that are capable of generating secondary events. Early warning, the behavior of operators, communication failures and bureaucratic conflicts are, they suggest, the areas in which further research could improve understanding of how cascading disasters propagate in critical infrastructure, and how their progress can be arrested. This approach to systemic risk is being actively pursued by the Centre for Risk Studies in Cambridge University [64].

1.3. Scenario planning

Four scenario role-playing exercise workshops were held with university students in Germany and France. These involved getting the students to dynamically ‘play’ through an imagined future. The game collapsed real time to highlight significant issues and to focus on people’s decision-making, emotional responses and imagined actions [65]. The scenario exercises were designed with input from 5 international experts with experience of using scenario planning in risk management, social science, economics and urban management. The design also drew on an extensive literature review, summarised as follows.

Various authors have suggested the use of scenario planning as a way of exploring the critical role of decision-making and spontaneous behavior in the stressful time of an emergency like a blackout [66–68]. Moats et al. [69] describe how, after major disasters, people have to make high consequence decisions with incomplete or inaccurate information under time pressure and suggest scenario planning as a way to prepare. Alexander [70] described how scenarios are used in emergency management training and how outcomes are used as the basis of crisis planning. Bradfield et al. [71] say public policy makers are increasingly using scenarios to involve agencies and stakeholders in policy decisions.

In the disaster management field, the scenario approach has been applied in ‘ShakeOut’ drills to raise public awareness [72,73] and various authors describe how they are used in emergency management [67,71,74–78]. In Germany, scenario planning was used to explore the impact of a blackout on health care [79]. An article by Envision Tomorrow [80] suggested scenarios could be of two basic types – ‘normative’, which describe a preferred outcome, and ‘exploratory’, which describe unknown futures. The scenario could be timed for pre, during or post-disaster. The blackout scenario in this research was an exploratory scenario timed for during the event.

In summary, in terms of a conceptual framework to guide the research, this brief literature review points to the following issues. How do people imagine events might cascade after a major blackout and produce a chain of consequences? How do people’s feelings change over the 72-h scenario? Are their signs of emergent groupings, coping strategies and adaptive behavior? How might communication failure impact on individual and communities?

2. Methodology

The aims of the scenario were:

1. To explore the potential cascading effects of a 3-day blackout and the implications for different types of people,
2. To learn how people might behave, react and feel in a blackout,
3. To gauge societal resilience to a relatively long power outage, and in the process to learn about preparedness and response to a possible crisis of a 3-day blackout in Europe.

The 54 participants at the workshops were university masters students who had no advanced knowledge of the nature or content of the scenario. Between 10 and 16 students attended each of the four workshops: two in Karlsruhe, Germany and two in Paris, France. To encourage more lively discussions the participants were divided into tables of 4 students playing the 4 roles: an elderly 80 year-old person in a wheelchair, a parent with 2 children, a single university student and a 12 year-old child, which to cover a range of age differences, different responsibilities, different concerns, different health conditions, and different flexibilities. The roles were assigned, as much as possible, according to the students’ background and experience. They were asked if they had experience of living with elderly or disabled people or if they had looked after younger siblings or were married or lived with a partner and the roles were assigned accordingly. The remaining students were asked to play themselves as university students. A mentor was assigned to each group to provide support during the scenario. In order to reduce the possibility of bias, the mentors were instructed that their task was to motivate and encourage the participants to think for themselves and to imagine how they would feel and react in their assigned role, but not to lead the group or provide suggestions or ideas. The mentors kept a note of the participants’ emotional reactions, their discussion and imagined actions during each time period.

The workshop lasted about 3 h and was divided into 4 half hour time periods representing different intervals during the blackout: the first 30 min; the next 8 h; 24 h and 3 days. There was a short break between second and third interval. With the addition of the first 30 min, these time periods were similar to those used in the scenarios in the Crisis Management Handbook about power outages in Baden-Württemberg by the Ministry of the Interior, Digitisation and Migration and Federal Office of Civil Protection and Disaster Assistance in Germany [81].

The participants were told that they were living in either Paris or Karlsruhe and “it is a warm day in the middle of summer in the middle of the week. It is 12:30 lunchtime and the electricity goes off”. They were told the reason of the blackout was unknown, that it was unclear when the electricity would come back or how widely the problem extended. Cascading impacts, largely drawn from the Crisis Management Handbook mentioned earlier [81], were pinned on the board: no traffic lights, no subway or tram, no telephone landline, no payment by bank cards, no ATM cash, no fuel at petrol stations and, in most cases, no tap water or toilet flush. In groups they brainstormed what would happen and began to create a picture of their feelings and reactions, the decisions they would make, and their expectations of the authorities and local community. The participants were required to make notes about their feelings and behavior on their individual worksheets. In time period 2 new cascading events were pinned up: shops and supermarkets closed, no mobile network, no Internet, flight cancellations and no trains between cities. After a break, time periods 3–4 continued. No information about

the cause or extent of the blackout was given at any time, just a reassurance from the authorities that they were doing their best to fix the problem as soon as possible.

All the participants, without exception, took their roles seriously and were able to imagine the expanding crisis through group discussion. During each time period mentors completed forms documenting their perceptions of the participants' understanding and their expectations of help from the authorities, family, friends and neighbors. The role-playing scenario was followed by an open discussion to share the main lessons and key factors.

3. Results

The data from the 4 workshops included individual students' notes, mentors' notes and semantic questions for each time period, plus feedback from the end of workshop discussions. The worksheets were analysed for their semantic content and provide the main data source for this results section. Both qualitative and quantitative content analyses were used to code the data [82,83]. A detailed content analysis was made of the words and phrases used by the participants to describe their feelings and reactions, their expectations of and opinions about the emergency services and government, authorities and their imagined behavior and interaction with others.

3.1. Emotional response

As explained in the methodology section, the participants recorded their feelings, thoughts, reactions and expectations on individual worksheets. This allows us to trace changes in their emotions and behavior over the four time periods. At the start of the game all the participants were happy and laughing, and many of them did not take the blackout seriously. However, by the start of the second 8 h period no one was laughing, but they still did not expect scenario to continue much longer. By the third period, after 24 h of blackout, the students were heavily into their roles and were frustrated and beginning to get scared. The way they argued and discussed within their group changed completely from the first time period. By the fourth period, after 72 h of blackout, they were very anxious and afraid.

The mentors' notes show that in the first 30-min period most participants (71%) expected the electricity to come back on any second and the remainder (29%) estimated a maximum outage of 2 h. Even after 8 h, over 75% were still hopeful that the power would be back within an hour. By the fourth period (72 h of blackout) the majority of participants had given up any hope that electricity could come back at all and were prepared for the worst. There were some differences between the workshops in France and Germany. Less than 10% of French participants imagined that the electricity could come back, while in Germany 33% still had hope at 72 h.

Table 2 shows the emotionally charged words that the students used to describe their feelings in each time period grouped by role. These words were identified through content analysis of the students' and mentors' notes by noting any instance any word that might describe an emotion was used and were not pre-defined in any way. Interestingly, there was little difference in emotional self-descriptions between French and German participants. The most appropriate or differentiating single word to describe the emotion of each group, has been highlighted in bold. A number of things are striking about this table.

Table 2
Emotional reaction (most repeated words with emotional content in 4 workshops).

	30 Mins	8 Hours	24 Hours	72 Hours
Child	Happy, Calm	Bored , Puzzled, Worried	Afraid, Scared , Worried	Afraid , Scared, Worried
Students	Puzzled	Annoyed , Worried	Scared, Worried, Anxious, Impatient, Angry , Overwhelmed	Afraid, Scared , Angry
Parents	Puzzled, Worried	Annoyed, Worried, Anxious, In control	Scared, Worried, Anxious, Hopeless, In control, Brave	Worried, Stressed, In danger, Strong
Elderly	Calm	Worried, Anxious, Weak, In danger	Afraid, Scared, Worried, Overwhelmed , Stressed, Hopeful	Scared, In danger, Weak , afraid

- For all participants there is an escalation of emotional response over the 72 h, from calm puzzlement to worry or fear;
- the third period (24 h of blackout) shows the largest range of emotions in terms of the number of words used;
- participants in the child's role used less words to describe their emotions than participants in other roles
- as one might expect, students playing parents with young children showed greater control and emotional endurance than participants in other roles; the elderly quickly went from calm acceptance, when they imagine the blackout to be of short duration, to fear and feelings of being overwhelmed due to the health implications of the outage.

The emotional pattern recorded by the participants accords with the behavior observed by mentors. The participants' self-descriptions started to appear distressed and confused after a blackout duration of 24 h. In the subsequent period, some gave up waiting for help from the authorities and began to think for themselves, organize with others or take responsibility and initiate action. Others who were unable to fend for themselves – children, elderly and disabled – settled into a state of fear.

To explore emotional response in greater depth, for the workshops in Paris, a modified form of semantic differential was developed comprising 20 pairs of words in French and English. Table 3 shows the English version of this list.

These were based on words used by the participants at the German workshops and represented increasing intensity of emotions from happy/unhappy to able/overwhelmed. They were presented in random order and participants were asked to select all the words that described how they were feeling in each time period.

The first thing to note is that in the first half an hour the number of negative and positive words was roughly equal. Over time the number of negative words used then steadily increased from time period to time period (Fig. 1).

Plotting some of the most used words Fig. 2 shows how positive emotions remain fairly stable and declining only a little over time, for example by students reporting that they were feeling less calm and safe. Conversely, the number of negative words, reflecting negative emotions, for example feeling scared or overwhelmed, dramatically increase over time.

There were some noticeable differences between the different groups (Fig. 3). Over half the students playing elderly people imagined that they would be scared after 24 h compared to less than a quarter of those playing parents and children and only 13% playing students. In contrast at 24 h the most energized group to take action, for example in searching for information or trying to get food and water, were again the students' group: 88% of them felt energized to do something to remedy the situation. By 72 h, however, their enthusiasm had slumped a little. The

Table 3
Semantic differential words.

Calm	Afraid	Brave	Distrustful	Strong
Panicked	Pleased	Unprepared	Relaxed	Out of control
Scared	Cool	Puzzled	Prepared	Energized
Bored	Patient	Weak	Peaceful	In danger
Sure	Anxious	Composed	Annoyed	Unafraid
Trustful	angry	Able	Unhappy	Unexpected
Expected	Impatient	Happy	Aggressive	Incontrol
Shocked	Passive	Overwhelmed	Worried	Safe

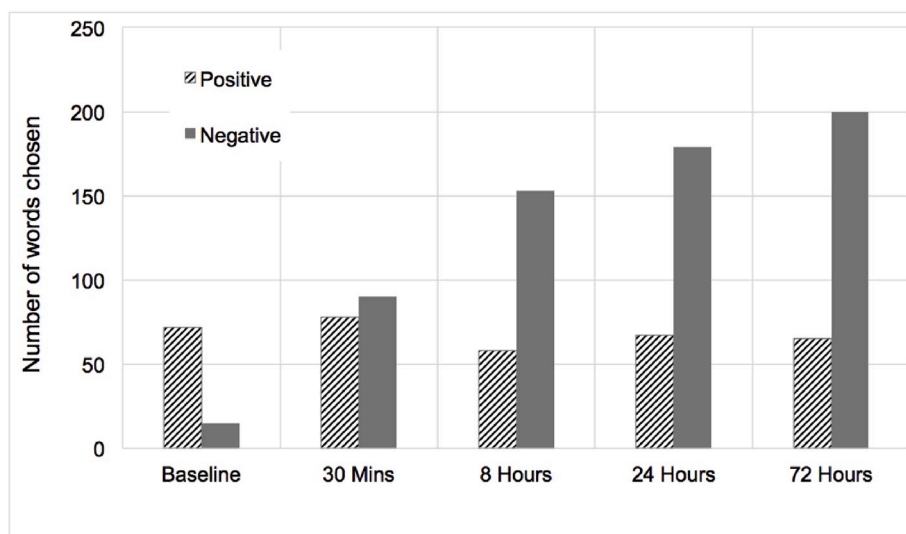


Fig. 1. Number of positive and negative words used over time.

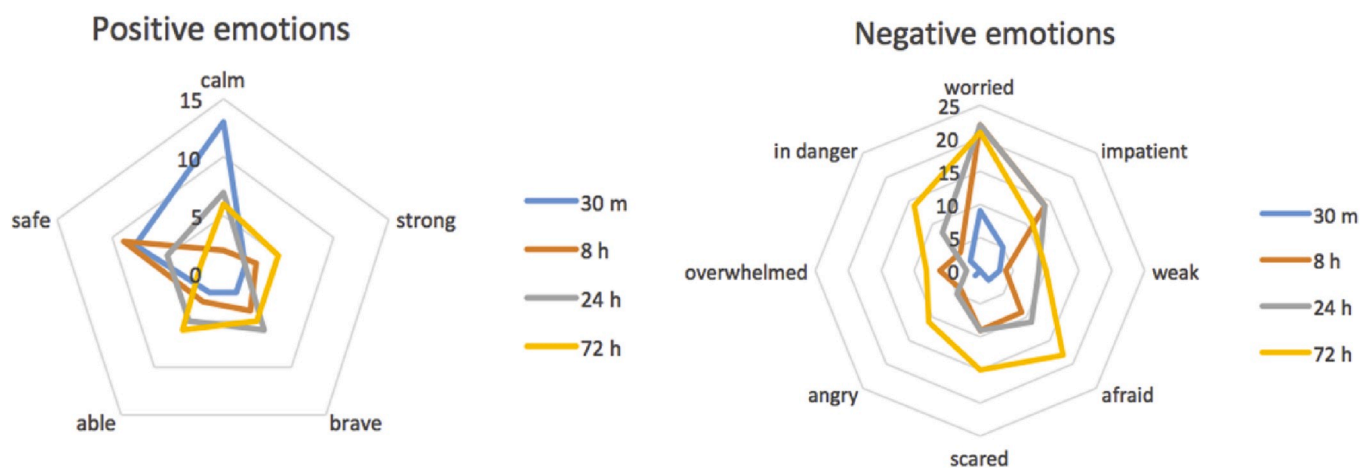


Fig. 2. Positive and negative words used over time (N = 31 participants).

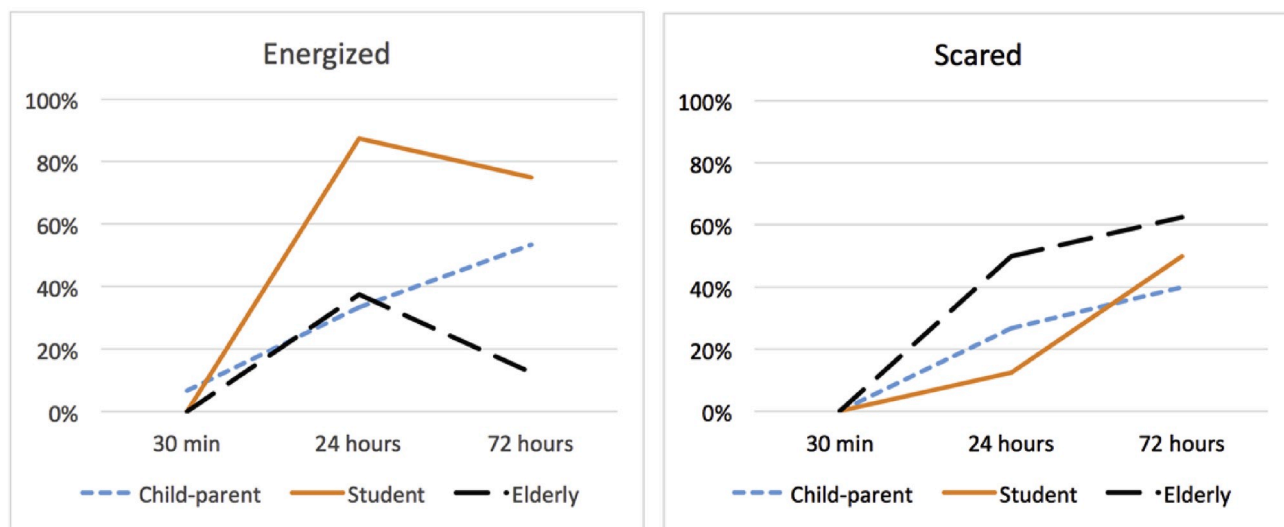


Fig. 3. Differences in emotion between groups.

energy levels of those playing parents steadily increased over the 72 h as they began to take charge of the situation and initiate action.

3.2. Behavior

As one might expect, there are certain patterns of behavior that are common to all four groups of participants and others that are specific. In the first 30-min period everyone wanted to get information about the cause and possible duration of the blackout. At 8-h over 80% of participants were still seeking information, the students in elderly roles were trying to contact people for help and those playing parents were beginning to anticipate a longer outage and to think about getting food and water. At 24-h those playing students were trying to find out how far the blackout extended to see if it made sense to evacuate. Those playing elderly were hoping to relocate to family or hospital. At 72-h people were getting increasingly concerned about security on the streets and some even appeared despairing. Interestingly, some stated their readiness to break into supermarkets to try to find food and water.

One of the most serious actions an individual or family experiencing a blackout could take would be to evacuate and leave their home and flee the area. As one would expect an expressed intention to evacuate increased dramatically over the four time periods to two-thirds of participants in Germany after 72 h and about one-third of participants in France (Fig. 4).

This difference between the participants in the two countries is arresting. It is possible that, unintentionally, the workshop leader had behaved differently in the German workshops and given a harsher impression of the lack of help from the authorities. However, a detailed analysis of the worksheets did not support this idea. The principal driver for evacuation would seem to be the lack of information, advice and support from the authorities and the need to find reliable supplies of drinking water and food. The city of Karlsruhe, where the German workshops were held, is of course much smaller than Paris where the French workshops were run. It is also nearer to an international border. There are differences in evacuation behavior between groups of people (Table 4). In both, Germany and France, a large proportion of “parents” and “students” begin to evacuate after 24 h. In Germany a very high proportion (80%) of “elderly” evacuate for health reasons when their family come from other cities in Germany by car to rescue them. In both France and Germany 72 h seems to be a critical time limit for the elderly beyond when they can no longer cope without medicine and health care and support. However, in all workshops, as the final decision to leave the region was made at a late stage and participants anticipated difficulties due to lack of fuel and traffic chaos.

3.3. People's expectations and trust in the authorities

Inaction by the authorities results in a loss of trust. Information plays a crucial role in an emergency and a failure in communication seems to be the most important reason for people losing trust in government. Everyone, no matter what age, gender, physical condition, or financial ability, needs to know what is happening in a crisis. A lack of information, which is very likely as reports of other major blackouts show, is a major problem for victims and dramatically increases uncertainty and anxiety. Uncertainty is an aversive state that people are motivated to reduce and is intensified by a lack of information [84]. The authorities often withheld information from the public in an effort to avoid panic [19]. Rather than helping the situation, communication failure can cause significant problems. Withholding information may delegitimize the government's authority and communication failure is thought to be responsible for people ignoring evacuation orders and other directives [85]. Communication is dramatically improved with social media and instant messaging through smart phones. Nevertheless, communication may be difficult or impossible during extended blackouts [86].

In this scenario, the participants were told that authorities were doing their best to find the cause of the blackout and fix the problem but are unable to say how long it will last. In open discussions, participants said that even though they would have been scared if they had been told it was a cyber-attack, they would have preferred to know rather than not. After 24 h, over half the participants started to react in an angry way (“still no information – that can't be true!”) and began to take things more and more into their own hands.

Apart from information about the cause of the event and about a probable duration, the participants expected advice from the authorities. The workshop participants initially trusted the authorities during the first two time periods, but with no guidance and information this trust dramatically dropped in the third time period. Fig. 5 illustrates the decline in trust compared to the express intention to evacuate. As a rule, people only evacuate when told to by the authorities, although the last 50 years there seems evidence that people's trust in government is declining and people place as much faith in advice from friends and family, often through social media connections as they do in government [87–91]. The findings shown in Fig. 5 may be an artifact of the scenario, in that the participants had no information about the blackout and no advice from the authorities, or they may point to a decline in trust.

The anticipated need for water increased strongly after 8 h and for food after 24 h for both, German and French participants. Both initially expected food and water to be provided by the authorities or by voluntary organizations. For the elderly, being cared for and being moved to a shelter after 24 h of blackout was considered the responsibility of the authorities.

Content qualitative analysis of the mentors' notes showed similar

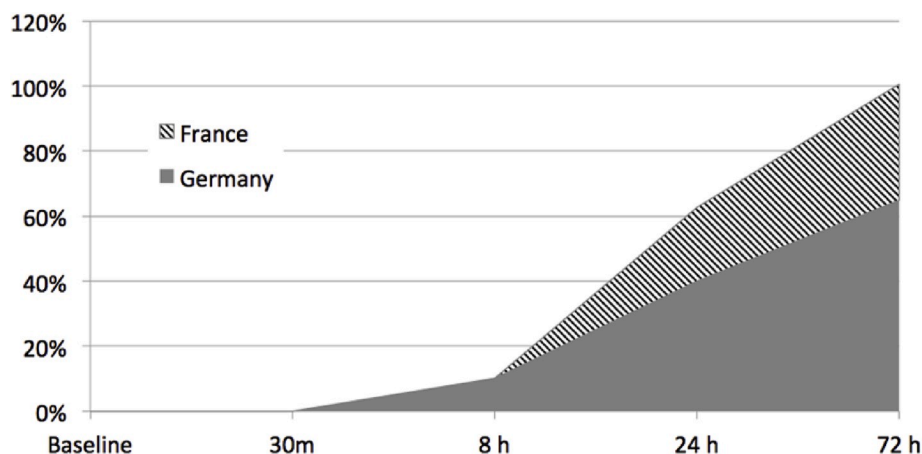


Fig. 4. Proportion of participants expressing an intention to evacuate.

Table 4
Intention to evacuate.

FRANCE	8 h	24 h	72 h	All	GERMANY	8 h	24 h	72 h	All
Child	0%	14%	0%	14%	Child	0%	25%	25%	50%
Student	0%	38%	25%	63%	Student	17%	33%	17%	67%
Parent	0%	38%	0%	38%	Parent	0%	40%	20%	60%
Elderly	0%	0%	25%	25%	Elderly	20%	20%	40%	80%
Total	0%	23%	13%	35%	Total	10%	30%	25%	65%

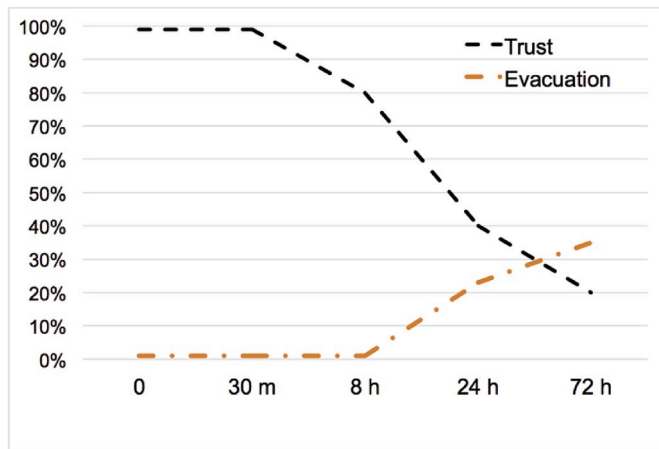


Fig. 5. Evolution of trust in authorities and propensity to evacuate.

expectations of government. As becomes clear from Table 5, the participants expected the authorities to provide information and increasing levels of assistance as the blackout continued. Since neither information nor support was forthcoming, participant's puzzlement turned to frustration, anger and finally fearful acceptance or action, first to self-organize and second to prepare for evacuation.

Table 5
Participants' expectations of the 'authorities' at different time periods.

30 Mins	8 Hours	24 Hours	72 Hours
- Information expected - The problem fixed soon - Short blackout - Confirm with people that public service is still functioning - Traffic police manage the traffic	- Information expected from police, Red Cross, government, city hall (Find the way to inform people considering people have no access to internet and TV) - Water distribution - Nursing care for elderly - Advice from government about how to behave and what to do if it continued - Provide water	- Information and guidance - Government provide food and water and other initial needs - Shelter provision for people in need - Help from other regions or other countries - Evacuation - Government visit old people at home - Employers to not expect their employees to go to work - Security force (police, army) provide security - Local government officials in the area - Local and central government implement emergency measures - University provides information - Community organizations step in - - Foreign authorities support foreigners - Provide portable toilets - Calm down people	- All points from previous time period plus: - Government plans organized evacuation - Firefighters provide humanitarian support

3.4. Volunteering and community cooperation

In both countries the level of volunteering and cooperation increased with each time period (Table 5). Volunteerism refers to the activities of people who work outside of formal emergency and disaster management arrangements to help others who are at risk or are affected by emergencies and disasters [92]. In Germany, however, volunteers are highly organized through the Bundesanstalt Technisches Hilfswerk [93] and there are over a million volunteer fire fighters and many relief organizations and local NGOs. In France, volunteering, although almost as important in terms of people involved as a percentage of the workforce, is less professionally organized and more ad hoc than in Germany [94, 95].

The most active group in the game, in terms of volunteering, were those playing the role of students. This phenomenon is similarly evident for other disasters in university cities and a common feature of altruistic behavior in disaster situations [96]. For example, during the 2013 flood in Germany, Passau was one of the most badly affected cities with 13 m of flood followed by 4 days of blackout in some parts of city. The first author of this paper interviewed 30 flood victims about the situation after flood and all respondents emphasized the positive role of university students in helping and supporting the flood victims for a full week. Interestingly, in the blackout scenario exercise, the same pattern emerged. Participants playing students mentioned they would help others, including strangers, from 8 h until the end of the blackout.

Participants playing the role of parents imagined helping and cooperating only with people they knew i.e. friends and neighbors. In

general, neighbors were considered to be the primary and most reliable source of information and support. Some old people suggested community cooperation would happen naturally, and that people would come together spontaneously with community leaders to plan and organize, for example to share food and support each other. Especially noteworthy is the spontaneous self-organization of family members and neighbors, which seem to build an autarkic “small-tie-society” on their own.

4. Discussion

In Europe, a blackout of 72 h is a very rare event. This makes studying and understanding an unprecedented crisis a challenge. Nevertheless, authorities and governments need to be prepared for such an event since the possibility of a blackout due to a cyber-attack, human error, or natural disaster is increasing and will eventually have huge negative effects on society. Authorities need to have an idea about people's potential reactions, their perceptions of the situation and their expectations of the authorities.

The participants in the workshops were a ‘convenience’ sample and none of the participants had experienced a long lasting blackout. The reported findings are the outcome of role-playing an imaginary situation by groups of young adults, mainly University Masters students, rather than the actual emotions and reactions of children, families and elderly people in a real blackout. Subjecting a real community to an actual 72 blackout to see what might happen would clearly have been unrealistic. This raises important questions of validity and reliability. The key questions are how valid and credible is the imagined communication blackout and how valid and reliable are the findings and specifically, what confidence can we have that the imagined behavior approximates reality?

In relation to the first question, an extended outage would impose severe strains on government and society as a whole [97,98]. Given our reliance on electricity to power our communications networks it is not unreasonable to assume that national and local government might have difficulty in informing citizens a few hours into the outage [99]. Government business would also be paralysed to an extent. It is possible to imagine a situation in which the authorities did not want to reveal the cause of the blackout for security reasons [100]. In this context scenario planning is a useful technique to anticipate events, which are mainly driven by people's behavior and beliefs. Using scenario planning in the way described in this paper is one of the ways one can begin to study, albeit hypothetically, people's reactions and behavior in order gauge societal resilience in such a disaster. Scenarios are descriptions of possible events that may occur in the future [101] and are a tool for exploring how new emerging risks could result in loss [102]. Gordon [103] in his review of futures research suggests that mark of excellence in a scenario is its internal self-consistency.

In relation to the second question, all the students were adults; at least half of them were Masters students, some with families, and the rest were third-year undergraduates. They adopted a very mature and engaged attitude to their roles and their imagined reactions and opinions are relevant and useful. One of most interesting aspects of the scenario was seeing students totally immersed in their hypothetical roles and imaginary situation. All the students took the game and their roles seriously and none dropped out. Each developed their own individual story, which together provide a fairly coherent picture of how a community hit by a major blackout might react. Albeit fictitious, the evolution of emotions, perceptions and motivations together with the provided explanations were highly plausible. Nevertheless, the scenario gave rise to important insights both in terms of material welfare and, more importantly, individual mental wellbeing and societal preparedness. Previous research suggests that even when people are aware of a natural or technological threat, this awareness does not necessarily translate into action [104].

Role-playing is a common technique used in studies of behavior in

which researchers assign research participants to particular roles and instruct those participants to act as if a set of conditions were true [105]. The technique capitalizes on people's ability to engage in creative play, to implement certain rules of thought, feeling, and behavior under certain assumed conditions. That students might be effective at playing arbitrarily assigned roles is bolstered by Goffman's [106] ideas about all social behavior being a form of dramatic performance. There is a considerable body of literature about using students as research subjects [107,108]. Two reasons are given for concern about using students: validity and generalisability and ethical issues relating to unfair coercion of students as subjects. Convenience was the underlying reason for using students as research subjects in this scenario exercises but the authors do not believe that there were any significant ethical issues as their presence was entirely voluntary and the research did not harm the people taking part nor intrude on their privacy nor threaten their beliefs [109]. Hanel and Vione [110] address the question of how far students can be representative of the general population. Using datasets of psychological variables the authors found that across 59 countries students are as heterogeneous as the general population and that their attitudes are as varied. Peterson and Merunka [111] could find no convincing empirical evidence of the negative consequences of using students for research or of the benefits, other than cost and convenience. Besides psychology, experimental economic research is one of the disciplines that predominantly use students as test subjects. The representativeness of student participants is also a regular point of criticism here, especially in contexts in which job-specific decisions are involved (e.g. farmers, investors). Again, comparisons show surprisingly little difference between student participants and professional representatives [112,113]. However, it would be interesting to repeat the exercise with carefully constituted focus groups of citizens and also possibly to extend the approach to places that experience frequent blackouts where people have evolved strategies to deal with the inconvenience.

Although the participants had the maximum of freedom to make their decisions it is interesting that the pattern of behavior followed a rather stable sequence. During the first 8 h waiting for information is the predominant reaction and most participants do not immediately interrupt their daily routine. These 8 h can be seen as a temporal buffer, which provides authorities the chance to fix the problem without drastic effects on an urban working day. However, if authorities cannot fix the problem and do not communicate with the population then people start taking matters into their own hands. In particular, the assumed lack of public communication was crucial. Although the situation was fictitious and the participants perceived the game as an entertaining role play most participants were genuinely upset to the government's failure to communicate. They expected information about the root cause of the problem and about the probable duration of the blackout. The predominant reaction of the participants, in particular of those in the role of parents who need to make the decision for the family, was understandable. They got together with neighbors and started to self-organize the provision of the necessities such as food, water, security, and medical support for the elderly and calming distraction for the children. At this point of time, they clearly stated that it was no longer a question of complying with the law and as soon as you believe you are left to yourself, you no longer feel obliged to obey the law. This finding has important implications for the communication policies of the public authorities in such a situation [114]. Although disaster victims may be very concerned and frightened and panic flight behavior can occur, it is quite rare and people usually behave rationally in weighing alternative course of action [115]. This behavior was noted in the students, who at a rather late stage, began to objectively and dispassionately discuss their options and only began to self-evacuate as a last resort. This behavior was restricted to the students playing the role single young adults and only in organized groups. They were quite realistic about the difficulties they would face if many people make this decision at the same time in a situation where they could no longer rely upon critical infrastructure.

It is interesting to speculate in what sense the psychological impact

of an extended blackout might be different to that of other disasters, such as storm and flood? And would Europeans who had never previously experienced an extended blackout react differently to people living in countries like Pakistan, Bangladesh, Iraq and Nigeria where blackouts are a common occurrence. Coping with crises is easier when you have the capacity to anticipate and cope with the resulting stress of an emergency such as a major power outage [116]. A study of electricity outages in Finland suggests two contrasting behavioral responses: on the one hand, the uncertainty imposed by a blackout can render people helpless while on the other uncertainty can be seen as a part of everyday life [117]. However, the study did not distinguish between different lengths of outage or whether information about its cause and duration was available or not.

The findings of what mature university students imagined would happen revealed that over the 4 time periods the students' emotional reactions went from frustration, frustration that their smart phones ran out of power or they couldn't get a signal, to worry about family, to anger with the authorities and finally to genuine life-threatening fear. This escalating emotional response was not only a product of inconvenience but also, and perhaps to a greater extent, due to the uncertainty and lack of information. What caused participants the most distress was uncertainty about how long the outage would last and a lack of information about the cause of the outage, how extensive it was and what was being done to fix it. Not knowing why the blackout was happening or how long it would last was the worst aspect of the experience for most of the workshop participants. This is quite different to other types of natural and man-made disasters, such as earthquakes, which happen suddenly and leave a devastating effect, or bush fires where the cause is known and understood. They can also see what firefighters are doing to help. This is not to suggest of course that the severity of these disasters is comparable in terms of loss and damage, but the aspects of uncertainty and lack of information are important considerations in European blackouts.

One way of assessing the reliability of these findings is by comparing them with the evidence reported in the literature. As reported earlier, after the New York blackout in 1965 people were initially annoyed and angry but later strangers were unusually helpful and friendly to each other [42]. And after the Ontario blackout in 2003 Murphy [44] reported that families had a variety of familial and social networks in their local area to call on for support. This is very similar to the behavior imagined by students in the scenario exercises. The impacts of the Ontario blackout reported by Anderson et al. [46] are also very similar to the cascading events generated by the scenario exercises. In the role play game various students said they would break into supermarkets to steal water and food. Although looting is rare in disasters it did occur during the New York blackout in 1977 [21]. A telephone survey of 2,000 German households [118] by Behrens GmbH Market Research in 2012 reported that 10% of respondents had experienced a long-lasting power failure. They found that in general people were largely unprepared for a major blackout; more than half the respondents has no information about power failures or felt a need to take precautionary measures. Perhaps, unsurprisingly, the more rural the respondents home area, the more self-sufficient they were in terms of food and water. The workshops reported here were conducted with people living in highly urbanized Karlsruhe and Paris and the availability drinking water and food were seen as serious immediate issues. The survey also found that there was no statistical effect on the level of preparation of age, household size, income, education or occupation. However, having a family member with health or disability issues did mean families had taken precautions to maintain life support systems and medical supplies.

5. Conclusion

Blackouts can cause major disruption and incur huge costs but their rarity poses particular problems for research. The study reported here used students as research subjects in France and Germany to role-play

the scenario of a 3-day blackout.

To return to the conceptual framework and the questions posed earlier. The students exhibited a clear understanding of how events might cascade after a major blackout and produce a chain of consequences, for example they imagined increased traffic jams and accidents as the result of traffic light failure, which expected would escalate if people could not contact the police because of mobile network or battery failure, they also imagined that hospitals might be overwhelmed because of an increase in accidents, such as setting fire to their home at night by dropping a candle or cooking on an open fire. How do people's feelings change over the 72-h scenario? Are their signs of emergent groupings, coping strategies and adaptive behavior? How might communication failure impact on individual and communities?

The findings show that people's emotions and perceptions change significantly over the 3 days from initial calm and an expectation that the blackout would be of short duration, through anger and frustration to desperation and genuine fear. A good proportion of the participants had, by the third period at 24 h duration, given up waiting for help from the authorities and were being to organize and look for solutions in terms of volunteering, cooperating with others, finding food and water and contemplating or initiating evacuation. Our findings suggest that people would move from frustration that their smart phones lost power or signal, through anger that the authorities were doing nothing to genuine anxiety and, in some cases, a fear for safety and life-support.

A study of water supply in Kathmandu, Nepal after the 2015 Gorkha earthquake showed that the chronic stressors, such as supply-demand disparity, could improve the adaptive capacity of both agencies and the public to deal with disruptions in infrastructure [119]. In the blackout scenarios there were clear signs of emergent groups, coping strategies and adaptive behavior, for example, people organizing into groups to take care of elderly, disabled and children, while other groups searched for food and information.

The results suggest that the main cause of the escalation in emotional response was not so much the inconvenience caused by the energy blackout, but the much more serious effects of the communication blackout. Not knowing what was happening or how long it would last caused the main distress. This suggests that the authorities need contingency plans not only to keep critical services supplied with power, but also how to communicate with citizens in a major extended power outage. To assist in preparing for infrequent crises such as major blackouts many observers have advocated that communities and governments engage in pre-disaster planning and preparation [120] and the more issues can be thought through in advance, including by means of scenarios, the greater will be the quality of crisis decision-making and management [121].

Although there are validity issues in using students as research subjects in role-play scenarios the findings from this research are plausible and have an inherent consistency and are indicative of how people would react in a real event that can inform policy. In particular authorities need to anticipate the anxiety occasioned by the information vacuum that would necessarily result from a major blackout. Two main messages are that people want to know what's happening, even if the news is bad, and secondly they want direction from the authorities.

Declaration of competing interest

□ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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